



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRM AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

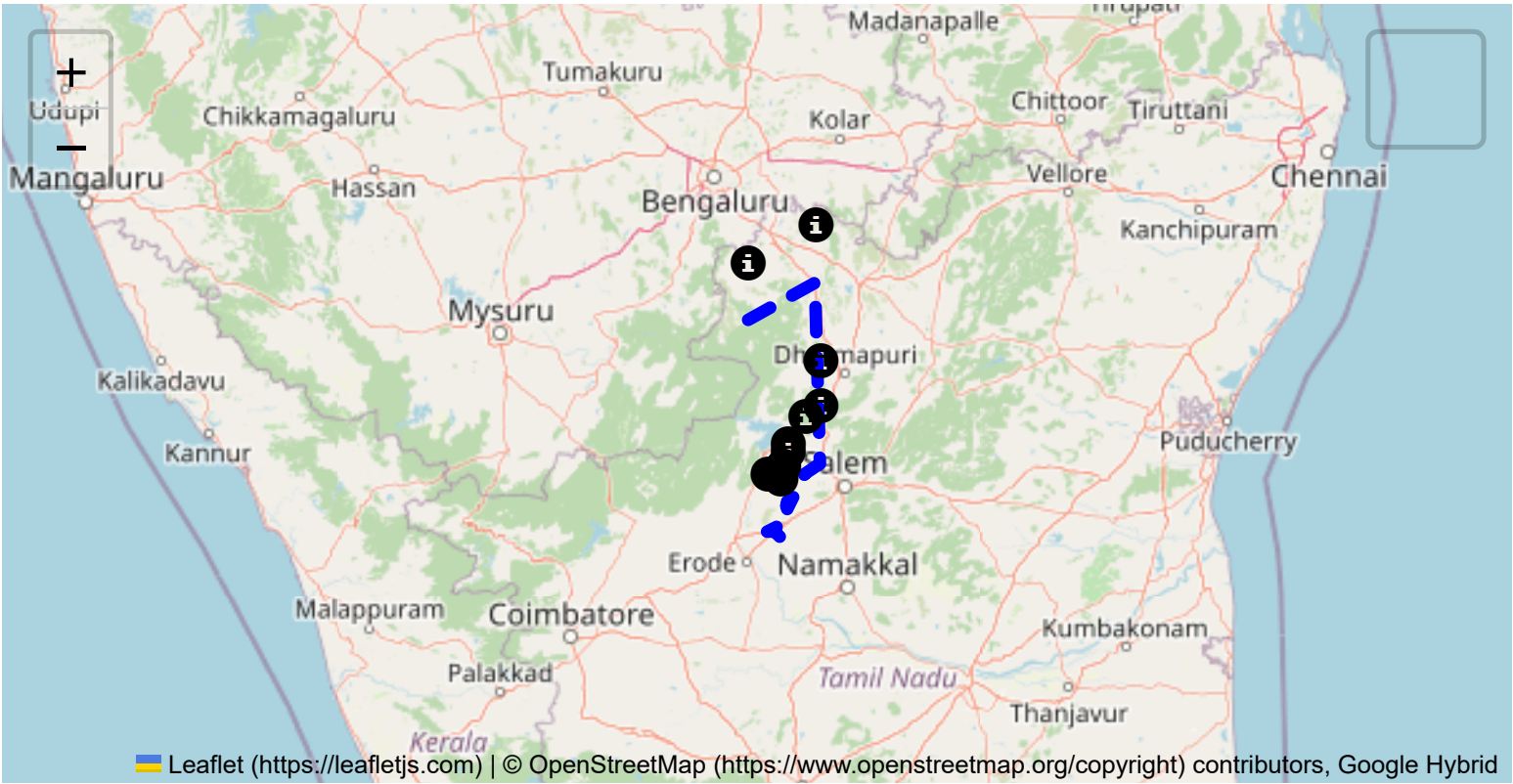
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

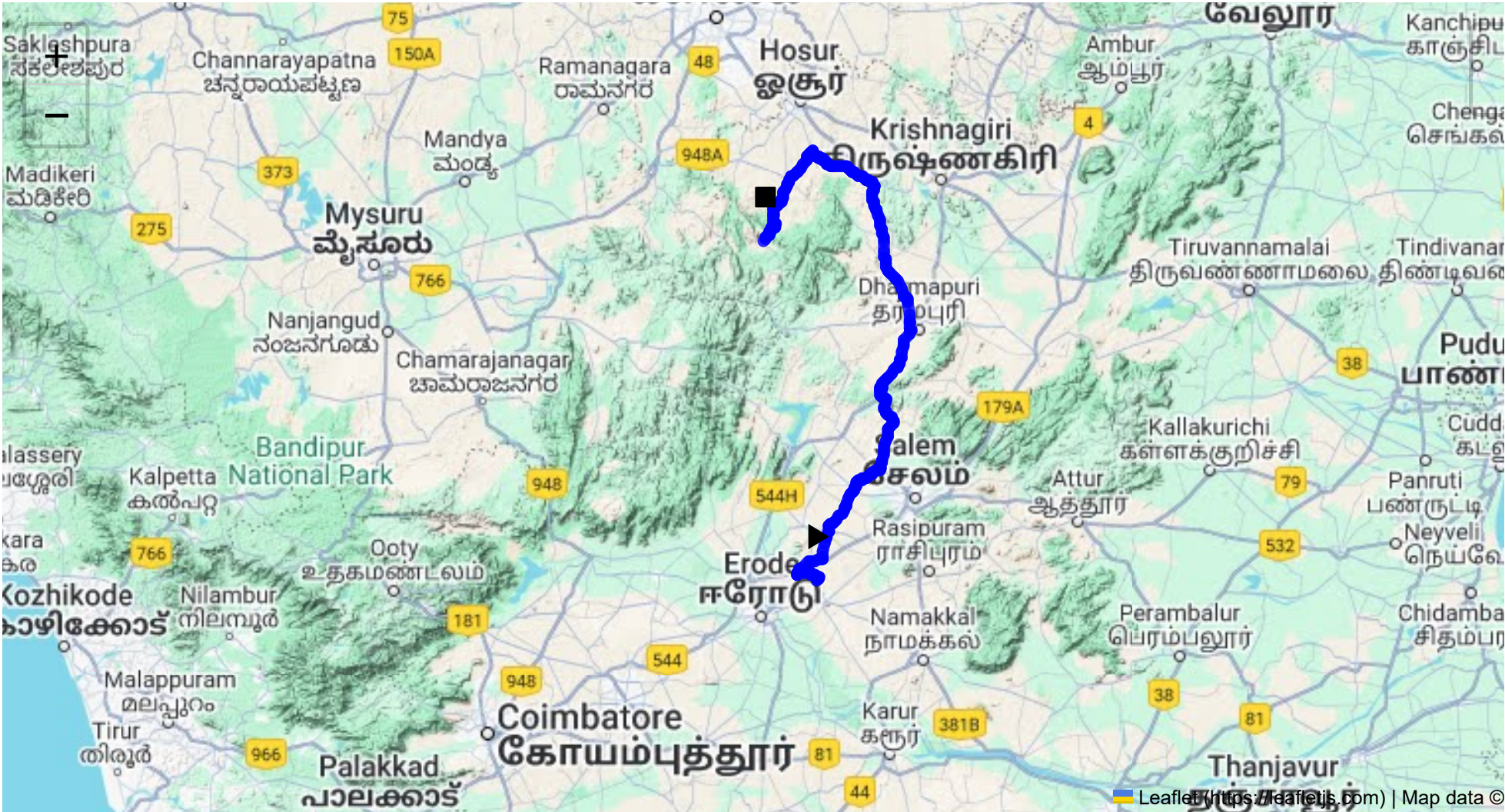
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 208.48 km
Estimated Duration: 4.4 hours
Adjusted Duration (Heavy Vehicle): 5.6 hours
Start: (11.4381, 77.8734)
End: (12.363324, 77.724921)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Hosur covers approximately 208.48 kilometers and involves various highways and state roads, including SH 86 and SH 85. The route traverses multiple towns and rural areas, requiring navigation through diverse terrains.

2. Weather Conditions and Potential Hazards

Tamil Nadu generally experiences a tropical climate, with hot summers, significant monsoon rainfall, and mild winters.

- **Monsoon (June to September and October to November):** Heavy rains can cause localized flooding and waterlogging, particularly in low-lying sections. Slippery roads and reduced visibility could be hazards.
- **Summer (March to May):** High temperatures may affect vehicle performance, especially tire wear and coolant levels.

3. Traffic Patterns and Congestion

- **Peak Hours:** Major congestion is often seen during morning (7:30 AM - 9:30 AM) and evening (5:00 PM - 7:30 PM) hours, especially near urban centers like Salem and Hosur.
- **Congestion-Prone Areas:** Market areas, junctions in larger towns, and stretches of road near Salem may experience slow-moving traffic.

4. Road Quality and Infrastructure

- **Road Quality:** Varies considerably along the route. National highways are generally in good condition, whereas some state roads may be narrow and less maintained.
- **Infrastructure:** Ensure attention at junctions and flyovers near larger towns due to complex road designs.

5. Alternative Routes for Emergencies

- **Alternative Pathway:** During emergencies, consider National Highways over state roads for bypassing smaller towns. Utilizing NH 44 for parts of the journey may provide faster and safer routing, despite potential extra distance.

6. Regulations for Hazardous Materials Transport

- **Local Regulations:** Tamil Nadu mandates strict adherence to regulations for transporting hazardous materials, including proper labeling, documentation, and safety gear.
- **Restrictions:** Certain urban areas might restrict heavy vehicle entry during peak hours.

7. Historical Incidents

- **Heavy Vehicle Incidents:** Past records indicate sporadic incidents involving heavy vehicles, often linked to overturning due to speed or load imbalance.
- **Hazardous Material Incidents:** Few recorded incidents with hazardous materials; however, risks of leaks or spills in congested zones remain a concern.

8. Environmental Considerations

- **Sensitive Areas:** Near Salem, recognized for its air quality concerns and natural preservations. Minimize air pollution and noise, especially during night shipping.
- **Wildlife Crossings:** Be aware of potential wildlife crossings in rural areas outside major towns.

9. Communication Coverage

- **Coverage:** Generally strong around urban centers and along major highways.
- **Dead Zones:** Potential for patchy mobile network coverage in rural or densely vegetated areas, particularly between smaller towns.

10. Emergency Response Times

- **Urban Centers (e.g., Salem, Hosur):** Faster response times, typically within 15-30 minutes due to proximity to emergency services.
- **Rural Areas:** Response times can extend beyond 45 minutes.

11. Overall Risk Assessment

- **Overall Safety:** Moderate risk level. Key risk factors include variable road quality, monsoon weather hazards, and traffic congestion in urban areas.
- **Mitigation Measures:** Regular vehicle checks, cautious drive through narrow roads, adhering to traffic norms, alternative routing during adverse weather, and emergency preparedness enhance safety.

In conclusion, this route requires careful navigation, with considerations for varied road qualities, potential adverse weather conditions, and traffic patterns. Adherence to local regulations and environmental considerations are crucial, alongside maintaining consistent communications for safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.60628, 77.86014	15 KM/Hr
1	U-Turn	High	11.7460396, 78.0473884	10 KM/Hr
2	U-Turn	High	11.7446053, 78.04729689999999	10 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44107, 77.87528	15 KM/Hr
5	Turn	High	11.44900, 77.87406	15 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	High	11.45571, 77.81702	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	Medium	11.45654, 77.81706	30 KM/Hr
10	Turn	Medium	11.46916, 77.82431	30 KM/Hr
11	Turn	High	11.49440, 77.88480	15 KM/Hr
12	Turn	Medium	11.57814, 77.90627	30 KM/Hr
13	Turn	Medium	11.59009, 77.90575	30 KM/Hr
14	Turn	Medium	11.59312, 77.90773	30 KM/Hr
15	Turn	Medium	11.62185, 77.94328	30 KM/Hr
16	Turn	Medium	11.63843, 77.94821	30 KM/Hr
17	Turn	High	11.64192, 77.95195	15 KM/Hr
18	Turn	Medium	11.64691, 77.95118	30 KM/Hr
19	Turn	Medium	11.68529, 77.97598	30 KM/Hr
20	Turn	High	11.70249, 77.97881	15 KM/Hr
21	Turn	Medium	11.71374, 77.99500	30 KM/Hr
22	Turn	Medium	11.72457, 78.01612	30 KM/Hr
23	Turn	Medium	11.73748, 78.03422	30 KM/Hr
24	Turn	Medium	11.73765, 78.03636	30 KM/Hr
25	Turn	High	11.73653, 78.03860	15 KM/Hr
26	Turn	Medium	11.74325, 78.04722	30 KM/Hr
27	Blind Spot	Blind Spot	11.74563, 78.04735	10 KM/Hr
28	Blind Spot	Blind Spot	11.74461, 78.04730	10 KM/Hr
29	Turn	Medium	11.77998, 78.05786	30 KM/Hr
30	Turn	Medium	11.86695, 78.08296	30 KM/Hr
31	Turn	Medium	11.91266, 78.05852	30 KM/Hr
32	Turn	Medium	11.93365, 78.06387	30 KM/Hr
33	Turn	Medium	11.93746, 78.05331	30 KM/Hr
34	Turn	Medium	11.94414, 78.04818	30 KM/Hr
35	Turn	Medium	11.95498, 78.04689	30 KM/Hr
36	Turn	High	11.95847, 78.04989	15 KM/Hr
37	Turn	High	11.96011, 78.04944	15 KM/Hr
38	Turn	Medium	11.96251, 78.05104	30 KM/Hr
39	Turn	High	11.96427, 78.05118	15 KM/Hr
40	Turn	High	11.96526, 78.04950	15 KM/Hr
41	Turn	Medium	12.12027, 78.13284	30 KM/Hr
42	Turn	Medium	12.28293, 78.06782	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
43	Blind Spot	Blind Spot	12.51393, 78.03186	10 KM/Hr
44	Turn	High	12.51379, 78.02933	15 KM/Hr
45	Turn	Medium	12.51721, 78.02557	30 KM/Hr
46	Turn	Medium	12.52269, 78.02290	30 KM/Hr
47	Turn	Medium	12.52651, 78.01023	30 KM/Hr
48	Turn	Medium	12.54273, 77.97854	30 KM/Hr
49	Turn	High	12.55108, 77.97248	15 KM/Hr
50	Turn	Medium	12.55286, 77.96596	30 KM/Hr
51	Turn	Medium	12.58211, 77.89398	30 KM/Hr
52	Turn	Medium	12.58519, 77.88178	30 KM/Hr
53	Turn	Medium	12.58894, 77.87870	30 KM/Hr
54	Turn	Medium	12.60497, 77.85468	30 KM/Hr
55	Turn	Medium	12.59351, 77.84812	30 KM/Hr
56	Turn	High	12.44998, 77.74933	15 KM/Hr
57	Turn	Medium	12.43471, 77.75040	30 KM/Hr
58	Turn	High	12.40964, 77.75703	15 KM/Hr
59	Turn	Medium	12.40303, 77.75564	30 KM/Hr
60	Turn	High	12.39768, 77.75235	15 KM/Hr
61	Turn	Medium	12.39813, 77.75148	30 KM/Hr
62	Turn	Medium	12.40109, 77.75023	30 KM/Hr
63	Blind Spot	Blind Spot	12.40272, 77.74836	10 KM/Hr
64	Turn	Medium	12.39460, 77.74346	30 KM/Hr
65	Turn	Medium	12.38452, 77.74085	30 KM/Hr
66	Turn	High	12.38309, 77.74141	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
15	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
16	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
18	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
19	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
24	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium
29	police	Denkanikottai Police Station	12.5208971, 77.7903372	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
17	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
28	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
30	school	Govt. Boys Hr. Sec School	12.5158402, 77.787696	30 km/h	Medium
31	school	Holy Cross Hr. Sec. School	12.5114655, 77.7867735	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 12.60628, 77.86014



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.7460396, 78.0473884



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.7446053, 78.04729689999999



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44107, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45654, 77.81706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46916, 77.82431



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49440, 77.88480



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57814, 77.90627



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59009, 77.90575



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59312, 77.90773



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62185, 77.94328



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63843, 77.94821



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64192, 77.95195



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64691, 77.95118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70249, 77.97881



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71374, 77.99500



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72457, 78.01612



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73748, 78.03422



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73765, 78.03636



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73653, 78.03860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74325, 78.04722



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.74563, 78.04735



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.74461, 78.04730



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86695, 78.08296



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93365, 78.06387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94414, 78.04818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95498, 78.04689



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95847, 78.04989



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96011, 78.04944



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96251, 78.05104



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96427, 78.05118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96526, 78.04950



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12027, 78.13284



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.28293, 78.06782



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51393, 78.03186



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.51379, 78.02933



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51721, 78.02557



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52269, 78.02290



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52651, 78.01023



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.54273, 77.97854



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.55108, 77.97248



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55286, 77.96596



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58211, 77.89398



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58519, 77.88178



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58894, 77.87870



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.60497, 77.85468



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59351, 77.84812



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.44998, 77.74933



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.43471, 77.75040



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.40964, 77.75703



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40303, 77.75564



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.39768, 77.75235



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39813, 77.75148



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40109, 77.75023



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.40272, 77.74836



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39460, 77.74346



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.38452, 77.74085



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.38309, 77.74141

